

Ranges and
Definitions

3 Steps to ABG
Interpretation

Apply steps 1&2:

Acidemia

Normal pH

Alkalemia

Compensation

CASES

Click these buttons to
REVEAL & HIDE
answers and graphics

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Normal PCO_2 is 40 ± 4
Respiratory acidosis - $\text{pCO}_2 > 44$
Respiratory alkalosis - $\text{pCO}_2 < 36$

Normal HCO_3 is 24 ± 2
Metabolic acidosis - $\text{HCO}_3 < 22$
Metabolic alkalosis - $\text{HCO}_3 > 26$

pH

PCO_2

PO_2

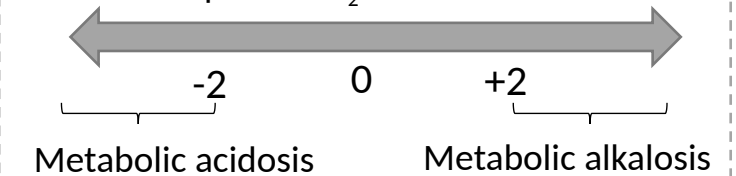
HCO_3

Normal pH is 7.40 ± 0.05
Alkalemia - $\text{pH} > 7.45$
Acidemia - $\text{pH} < 7.35$

Normal PO_2 on an ABG is > 80 .
Check out or talk on hypoxemia [here](#).

Base excess

Some labs will report a "base excess". This is the # of H^+ ions required to restore normal pH if PCO_2 is normal.



- "-emia" refers to the pH
- "-osis" refers to a metabolic or respiratory process that would lead to a derangement in pH if unchecked

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Use the menu on the left-hand side of the page to navigate to each page

Step 1: Acid-Base status?

Acidemia pH
<7.35

Normal pH
7.35 – 7.45

Alkalemia
pH > 7.45

Primary process?

Metabolic? Respiratory? Or both?

Step 2: Anion gap metabolic acidosis?

An anion gap metabolic acidosis can be present even with a primary respiratory process or primary metabolic alkalosis.

If yes, is there a concurrent metabolic process? (Delta-delta)

Step 3: Assess for compensation

Acidemia pH < 7.35

Step 1

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Step 2

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Metabolic
 $\text{HCO}_3^- < 22$

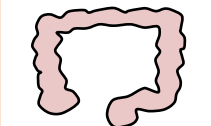
↑ Anion gap (AG)?

No

Yes

Urinary AG

NeGUTive



GI loss of HCO_3^-

Diarrhea
Ileostomy

Positive



Renal loss of HCO_3^-

↑ Cl^- (excess NS)
RTA
Early renal failure
↓ aldosterone

$\Delta\Delta$ (delta-delta) or "residual HCO_3^- "
Is there another metabolic process?

<1

1

>2

AGMA +
NAGMA

Pure
AGMA

AGMA +
metabolic
alkalosis

Methanol
Uremia
DKA (or any ketoacidosis)
Paraldehyde
INH (isoniazid), Iron
Lactic acidosis
Ethylene glycol and EtOH
Salicylates

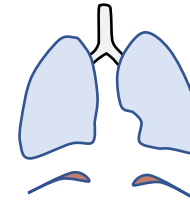
Respiratory
 $\text{pCO}_2 > 44$

↓ CO_2
elimination

Obstructive (and intrinsic)
lung disease
Neuromuscular weakness
↓ of RR (opioids, benzos)

↑ CO_2
production

Sepsis
Overfeeding
Hyperthyroidism



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Step 1

Step 2

Normal pH

Look for metabolic (HCO_3^-) or
respiratory (PCO_2) derangements
that might suggest mixed acid-base
disorder

↑ Anion gap (AG)?

Yes

$\Delta\Delta$ (delta-delta) or
“residual HCO_3^- ”
Is there another metabolic process?

Mixed AGMA
+ NAGMA

Pure AGMA

Mixed AGMA +
metabolic
alkalosis

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Apply steps 1&2:

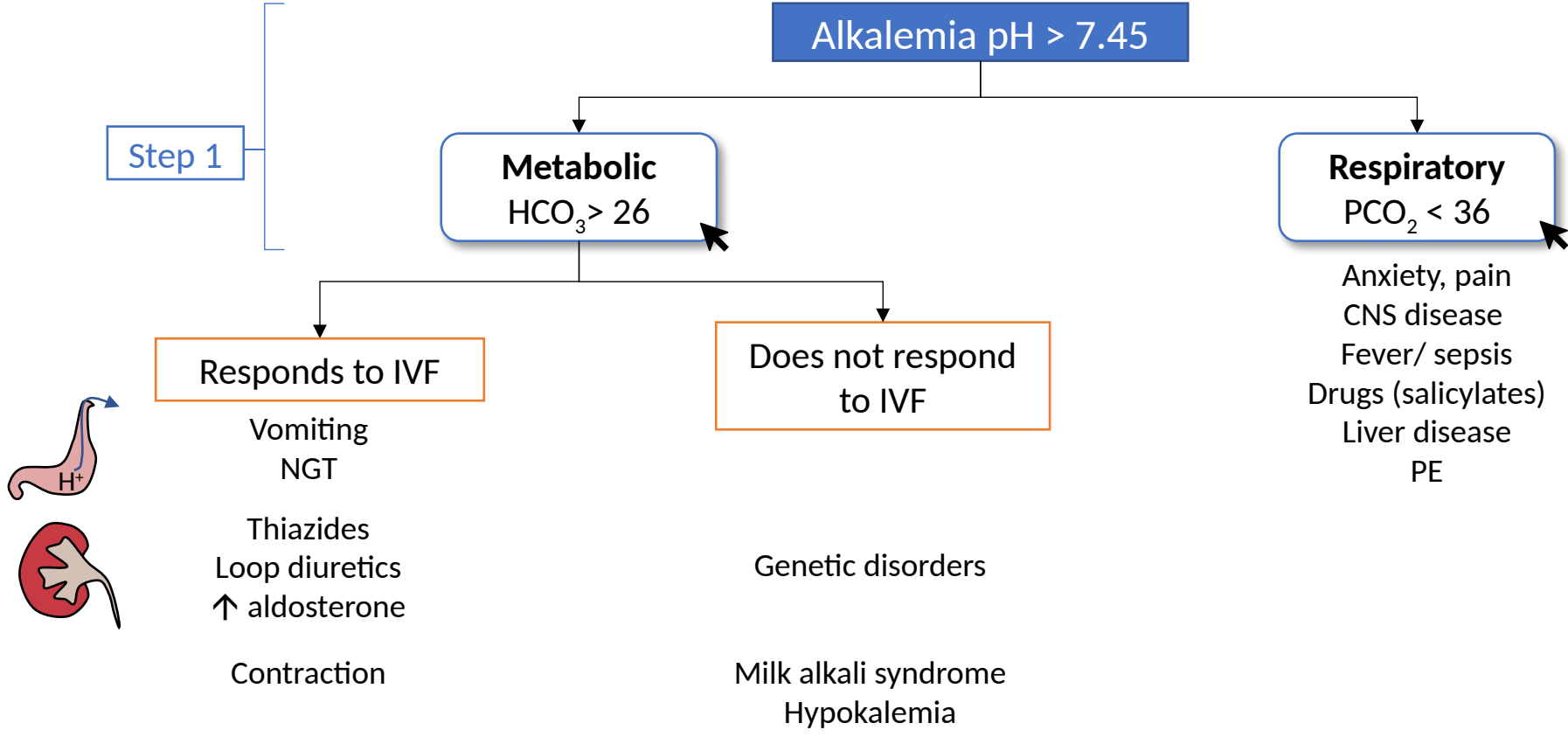
Acidemia

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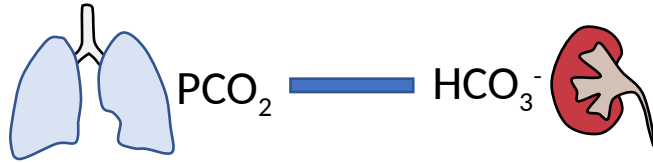
Alkalemia

Compensation

CASES

COMPENSATION

Step 3: COMPENSATION



The kidney and lungs help regulate pH balance. The HCO₃⁻ and PCO₂ move in the same direction.

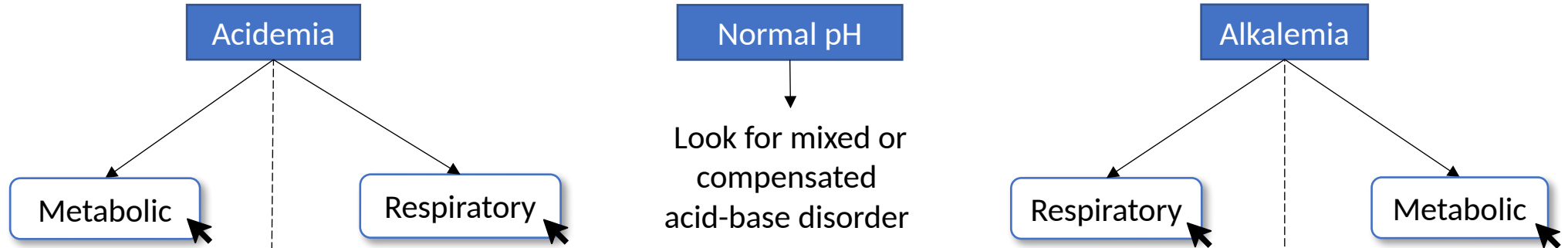
- When HCO₃⁻ is ↓ (metabolic alkalosis), the body will eliminate CO₂ (↓) to compensate.
- When PCO₂ is ↑ (respiratory acidosis), the body will retain HCO₃⁻ (↑) to compensate.

Respiratory (PCO₂) compensation occurs rapidly (minutes).

Metabolic (HCO₃⁻) compensation occurs more slowly (2-3 days).

Compensation will never overcompensate or fully correct pH to normal (7.40).

PRIMARY PROCESS



Respiratory alkalosis

1. Winter's formula
 $PCO_2 = (1.5 \times HCO_3^-) + 8$

2. Thumb rule
PCO₂ should be ≈ to last digits of pH
↓ 32/30/-/18



Metabolic alkalosis

ACUTE: ↑ HCO₃ 1

CHRONIC: ↑ HCO₃ 4

40
PCO₂

Metabolic acidosis

ACUTE: ↓ HCO₃ 2

CHRONIC: ↓ HCO₃ 5

Respiratory acidosis

$PCO_2 = (0.7 \times HCO_3^-) + 21$

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Basic

1-3

4-5

Click each shaded box to reveal the answer.

↓pH / ↓PCO₂ / PO₂ / ↓HCO₃

Acidemia

Metabolic acidosis

↓ PCO₂

↓pH / ↑PCO₂ / PO₂ / ↓HCO₃

Acidemia

Concurrent metabolic and respiratory acidosis

-

↓pH / ↑PCO₂ / PO₂ / ↑HCO₃

Acidemia

Respiratory acidosis

↑ HCO₃

↑pH / ↑PCO₂ / PO₂ / ↑HCO₃

Alkalemia

Metabolic alkalosis

↑ PCO₂

↑pH / ↑PCO₂ / PO₂ / ↓HCO₃

Alkalemia

Respiratory alkalosis

↓ HCO₃

↑pH / ↓PCO₂ / PO₂ / ↑HCO₃

Alkalemia

Concurrent metabolic and respiratory alkalosis

-

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Case 1

34F is found down. In the ED, her SaO₂ was 89% on RA. Pinpoint pupils on exam. ABG: 7.25/60/60/26, base excess of 1.

136	100		Albumin 4.0
	26		

Acidemia. pH of 7.25 is low.

Respiratory acidosis. PCO₂ is ↑ (nml 36-44) driving the acidemia. HCO₃ is at the upper range of normal.

There is no concurrent AGMA.
(Na) - (HCO₃ + Cl) = 136 - (100 + 26)
AG = 10 (normal 8-12)

No ΔΔ needed.

None. HCO₃ is nml. No base excess. For acute PCO₂ ↑ 20 → ↑ HCO₃ by 2 (26).

A-a difference is 15, normal for her age.

Acute respiratory acidosis and hypoxemia from hypoventilation likely from narcotic overdose.

Case 2

45M with IBD presents with abdominal pain and multiple loose bowel movements. ABG: 7.34/35/-/18.

127	100		Albumin 4.0
	18		

Acidemia. pH is borderline low.

Metabolic acidosis. Both PCO₂ and HCO₃ are ↓. Because the pH is borderline low, the metabolic acidosis is the 1° process.

There is no AGMA. This is a NAGMA.
(Na) - (HCO₃ + Cl) = 127 - (100 + 18)
AG = 9 (normal 8-12)

No ΔΔ needed. UAG would not be accurate given renal dysfunction.

Yes. Resp alkalosis. Expected PCO₂ by Winter's formula = (1.5 x HCO₃) + 8 = 35. The expected PCO₂ by the thumb rule is also 35.

Non-gap metabolic acidosis with compensatory respiratory alkalosis from diarrhea.

Case 3

65M with COPD presents with fever and cough. His SaO₂ is 90% on 30% FiO₂ by Venturi mask. ABG: 7.36/60/60/36.

142	97		Albumin 3.0
	32		

pH is low-normal.

Respiratory acidosis. Both PCO₂ and HCO₃ are ↑. The pH is low-nml, thus, respiratory acidosis is the 1° process.

There is no AGMA.
(Na) - (HCO₃ + Cl) = 142 - (97 + 36) = 9
Expected AG = 2.5 x albumin = 7.5.

No ΔΔ needed.

Yes. Met alkalosis. pH is nml so likely chronic. For chronic PCO₂ ↑ 20 → ↑ HCO₃ by 8 (32).

A-a difference is 72, ↑ for his age.

Chronic, compensated resp acidosis. High A-a gradient hypoxemia may be related to VQ mismatch or shunt.

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4-5

Case 4

72 yo F with RYGB presents with 3 d of shortness of breath, abdominal pain, nausea and diarrhea. She received IV morphine for her pain and became somnolent. An ABG showed: 6.98/41/58/10.

146	121	18	98
	7	1.0	

Lipase 3290 Lactate 2.7
Albumin 2.1
Urine: + ketones

Acidemia. pH of 6.98 is low.

Metabolic acidosis (HCO_3 is $\downarrow$). PCO_2 is normal, but there is likely a concurrent primary respiratory process (see compensation).

There is an $\uparrow$ AG.
The AG = $146 - (121 + 7) = 18$.
The differential is MUDPILES.

There is a concurrent NAGMA (likely from diarrhea).
Using the "residual HCO_3 " method, the $\Delta\text{AG} = 18 - \text{expected AG}$. With an albumin of 2.1, the expected AG is 5-6. $\Delta\text{AG} = 12$.
Adding ΔAG to HCO_3 of 7 = 19, which is still a metabolic acidosis.

Using Winter's formula: Expected $\text{PCO}_2 = (1.5 \times \text{HCO}_3) + 8 = 19$.
Measured PCO_2 is 41, suggesting a concurrent respiratory acidosis.

There is AGMA from ketoacidosis and lactic acidosis, NAGMA likely from diarrhea, and a concurrent respiratory acidosis likely from narcotic associated hypoventilation.

Case 5

24 yo M with a history of depression is brought into the ED unresponsive. On exam, his RR is 34 breaths/min, SaO_2 99% on RA. ABG is 7.56/22/110/17.

137	99		
	18	1.2	

Albumin 4.0
Lactate 1.2
Urine: bland

Alkalemia. pH is high.

Respiratory alkalosis. PCO_2 is $\downarrow$.

There is an $\uparrow$ AG. AGMAs are never a compensatory process so there is a concurrent primary AGMA.
 $\text{AG} = 137 - (99 + 18) = 20$

This is a pure AGMA.
Using the "residual HCO_3 " method, the $\Delta\text{AG} = 20 - \text{expected AG (12)}$. $\Delta\text{AG} = 8$. If we add this to measured HCO_3 , the "residual HCO_3 " is 26, which is upper limit of normal.

None. There is a concurrent primary AGMA.

A primary respiratory alkalosis and AGMA is classic for salicylate poisoning.

Take-Home Points

1. Read every ABG the same way: (1) pH — acidemic, alkalemic, or normal; (2) primary process — match pH with PCO₂ (resp) and HCO₃ (metabolic); a normal pH with deranged PCO₂/HCO₃ means a mixed disorder.
2. Calculate an anion gap in EVERY patient, whatever the primary process: $AG = Na - (Cl + HCO_3)$; expected $AG \approx 2.5 \times \text{albumin}$. An AGMA can hide behind any primary disorder.
3. If AGMA: do the delta-delta ("residual HCO₃"): add (measured AG – expected AG) back to the HCO₃ — high → concurrent metabolic alkalosis; low → concurrent NAGMA; normal → pure AGMA.
4. Compensation moves PCO₂ and HCO₃ in the SAME direction; respiratory compensation is fast (minutes), metabolic slow (2–3 days); it never overshoots or fully normalizes pH. Metabolic acidosis → Winter's formula: $PCO_2 = 1.5 \times HCO_3 + 8$ ($\approx$ last two digits of pH).
5. Inadequate or excessive "compensation" = a second primary process. Mismatch between measured and expected values is a diagnosis, not a rounding error.

Attribution & Changelog

Original

"Acid Base Disorders" — TeachIM.org (Mar 2021). Author: Yilin Zhang MD; initial expert review Andy Luks MD (Pulm/ICU, UW).
https://teachim.org/teaching_material/acid-base-disorders/ — CC BY-NC 4.0 (educational, non-commercial use).

Corrections in this optimized copy (June 2026) — applied to BOTH boards

Basic-cases slide, bottom-right case ($\uparrow$ pH/ $\downarrow$ PCO₂/ $\uparrow$ HCO₃): "Concurrent metabolic and respiratory acidosis" → "...alkalosis" (label contradicted the case's own arrows).

Case 5: " Δ AG = 19 - expected AG" → "= 18" (the slide's own AG = $146 - (121 + 7) = 18$; Δ AG 12 unchanged).

Case 4: "24 yo M ... her RR" → "his RR" (pronoun/sex mismatch; flagged for author review).

Appended slides only — all original slides, graphics, and click-to-reveal animations preserved on both boards.

Currency

Framework verified June 2026: stepwise interpretation, AG with albumin correction (expected $\approx 2.5 \times$ albumin), delta-delta, and Winter's formula remain standard teaching — no guideline changes to incorporate.

Companion files

ABG-Learner.pptx (corrected) · Acid-Base-Worksheet.docx · Acid_Base_Lesson_Package.html · Acid_Base_OnePager.pdf · Acid_Base_Anki_Cards.txt